Flight Test Engineering

Section 5020

Module 1

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3 Model Configuration

Parameter	Value
K_V	420 $\frac{RPM}{V}$
Winding Resistance (per phase)	0.087 Ω
No load current	0.68 A
Maximum current	35 A
Maximum torque	0.79 Nm
Maximum power	771 W
$K_e \approx K_t$	0.0227 $\frac{Nm}{A}$

Table 1: Motor parameters

Parameter	Value
Q_{vec}	$[-0.0101\ 0\ 0.0001\ 0\ 0\ \ -0.001\ 0\ 0\ 0\ 0]$
Mass	0.045 kg
Diameter	0.4064 m
Radius	0.2032 m

Table 2: Propeller parameters

Parameter	Value
Consistent energy capacity	115 Wh
Initial SOC	100%

Table 3: Battery parameters

6 Required Tuning Evidence

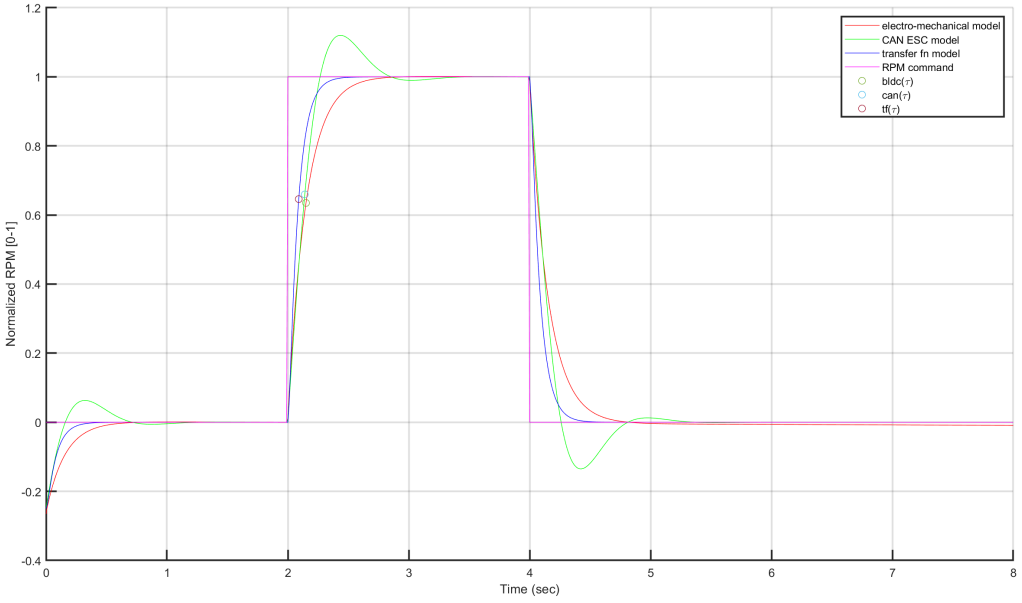


Figure 1: Normalized Overlay

The final values for the CAN ESC were

$$K_p = 0.1 \tag{1}$$

$$K_I = 1.0 \tag{2}$$

The mode shape of the electromechanical model and the 2nd order transfer function agree well, and both appear first order (despite the name of the latter). On the other hand the mode shape of the CAN ESC model is clearly second order, with a small overshoot of about 1/10th of the normalized RPM. The CAN ESC model was tuned to match the electromechanical model, and both have very similar rise times of 0.15 and 0.14 seconds respectively. The 2nd order transfer function is slightly faster, but in the normalized overlay all values at 1 time constant are nearly on top of each other. The steady-state tracking errors are also all very similar, with the highest error unsurprisingly coming from the electromechanical model, which is the only model that is feedforward and will not decay arbitrarily close to the commanded RPM. Still, an error of 8 RPM is on the order of 10^-3 of the steady state value, and should be regarded as essentially indistinguishable.

Model	**Time to rise**	**Steady-state error**
Electromechanical	0.15 sec	8 RPM
CAN ESC	0.14 sec	0.0721 RPM
Transfer function	0.09 sec	-6.6e-09

Table 4: Motor model differences

7 Comparisons

Parameter	**Value**
Averaging window (0)	1.8-2.0 seconds
Averaging window (f)	4.8-5.0 seconds
Observation interval	0.0-8.0 seconds

Table 5: Comparison parameters

7.1 1300 + 50 PWM

Model	10-90% **rise time (s)**	**Settling time (s)**
electromechanical	0.37	2.76
CAN ESC	0.14	2.97
transfer fn	0.17	2.36

Table 6: 1300-1350 PWM step response comparison

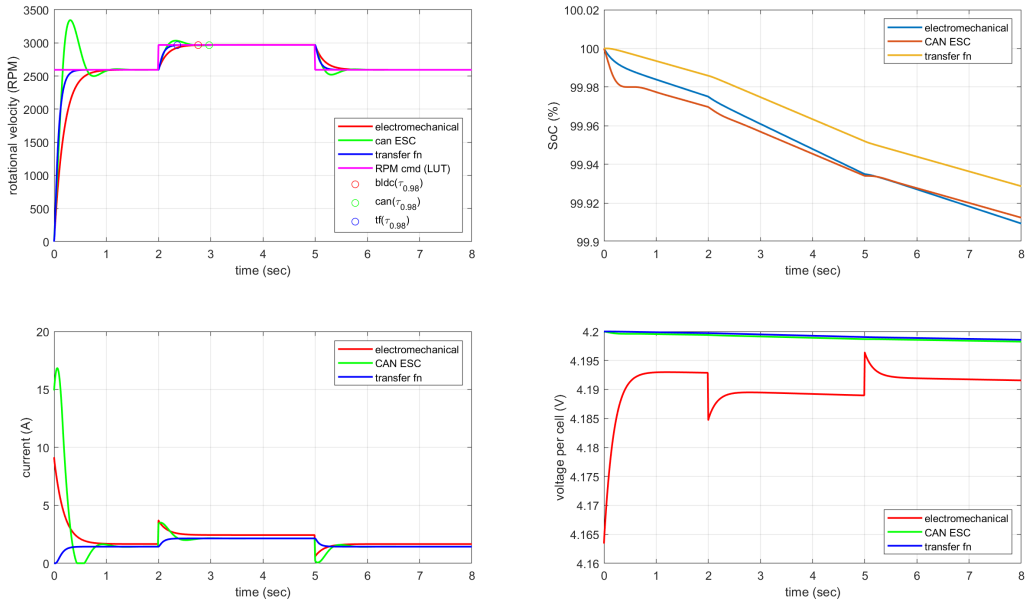


Figure 2: 1300-1350 PWM step response comparison

7.2 1300 + 150 PWM

Model	10-90% **rise time (s)**	**Settling time (s)**
electromechanical	0.35	2.72
CAN ESC	0.16	3.04
transfer fn	0.17	2.36

Table 7: 1300-1350 PWM step response comparison

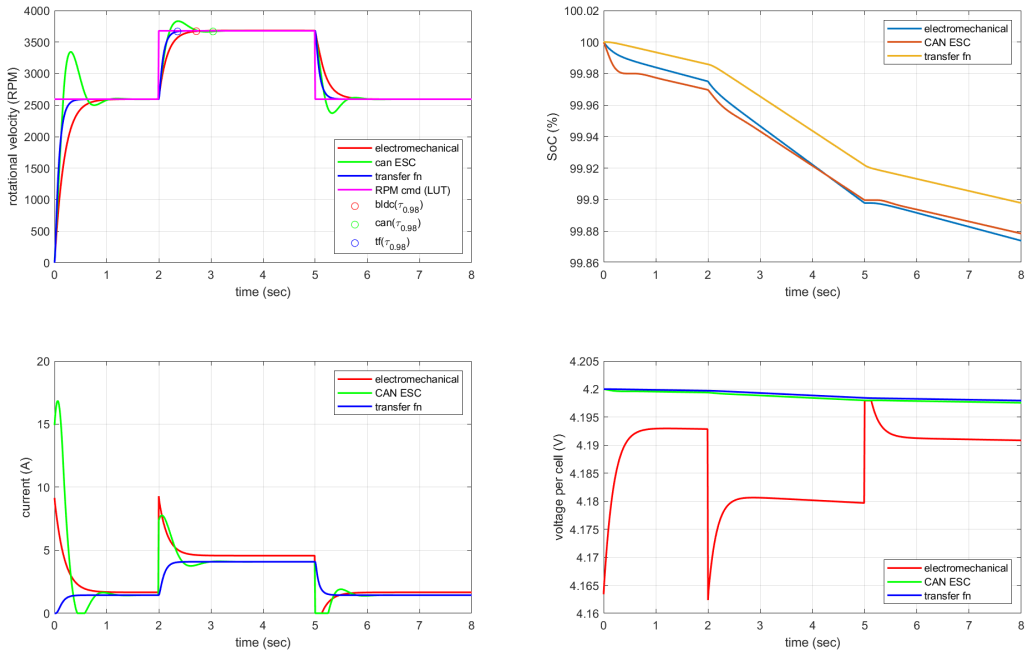


Figure 3: 1300-1450 PWM step response comparison

7.3 1300 + 300 PWM

Model	10-90% **rise time (s)**	**Settling time (s)**
electromechanical	0.33	2.67
CAN ESC	0.18	2.25
transfer fn	0.17	2.36

Table 8: 1300-1600 PWM step response comparison

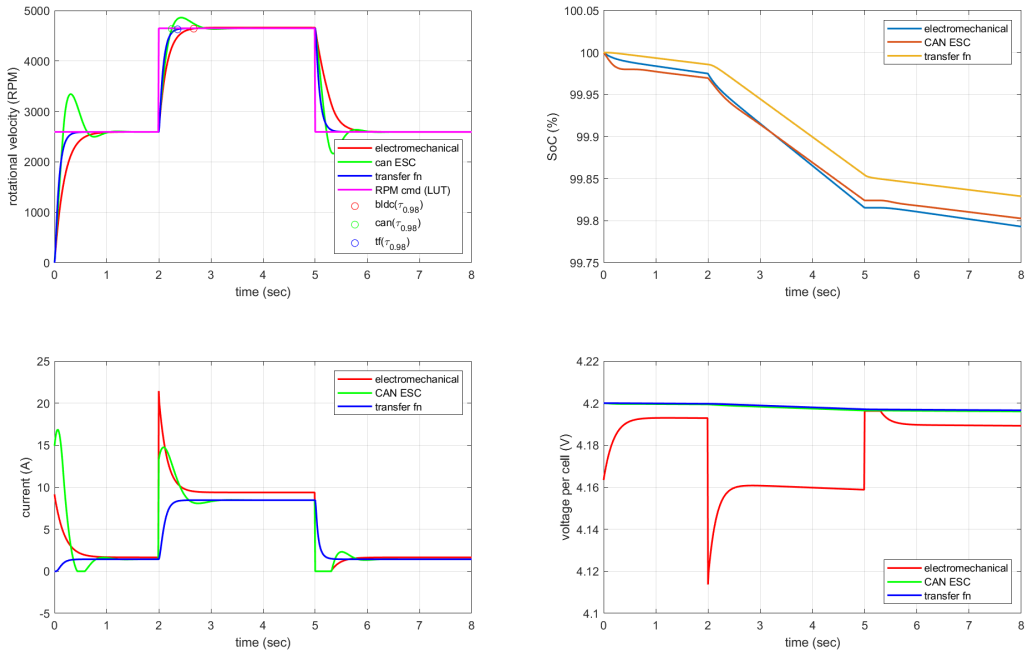


Figure 4: 1300-1600 PWM step response comparison

7.4 1500 + 50 PWM

Model	10-90% **rise time (s)**	**Settling time (s)**
electromechanical	0.32	2.63
CAN ESC	0.2	2.28
transfer fn	0.17	2.36

Table 9: 1500-1550 PWM step response comparison

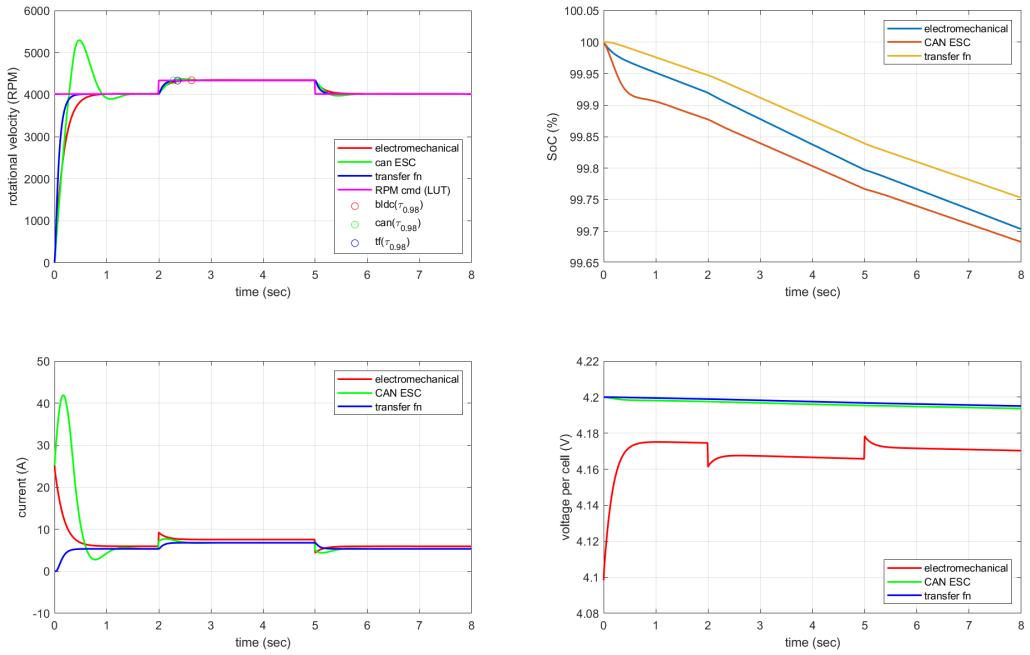


Figure 5: 1500-1550 PWM step response comparison

7.5 1500 + 150 PWM

Model	10-90% **rise time (s)**	**Settling time (s)**
electromechanical	0.31	2.63
CAN ESC	0.23	2.31
transfer fn	0.17	2.36

Table 10: 1500-1650 PWM step response comparison

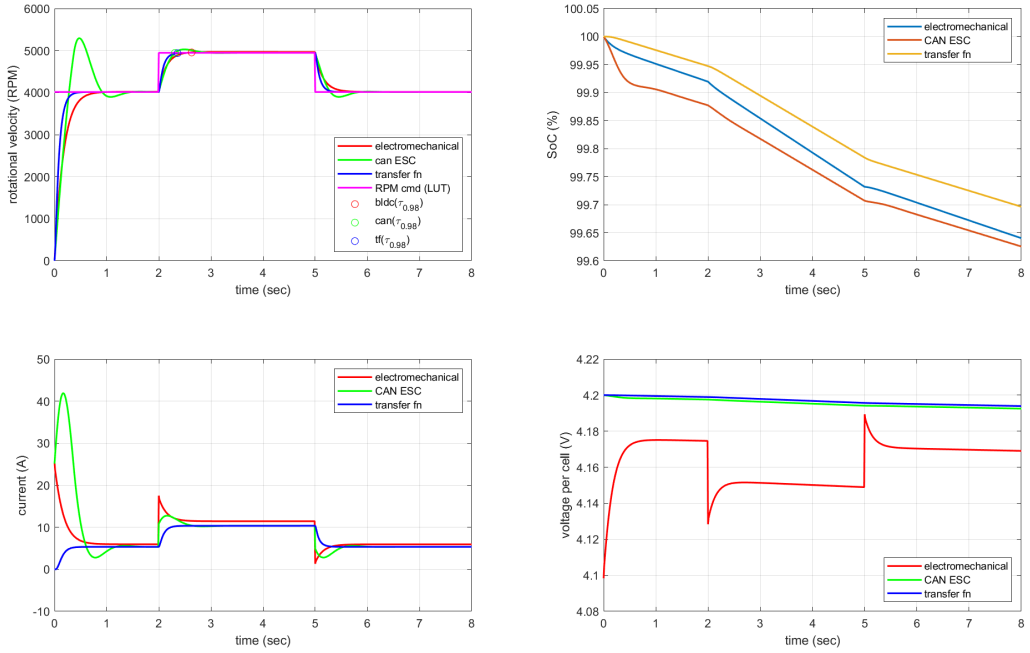


Figure 6: 1500-1650 PWM step response comparison

7.6 1500 + 300 PWM

Model	10-90% **rise time (s)**	**Settling time (s)**
electromechanical	0.3	2.6
CAN ESC	0.26	2.36
transfer fn	0.17	2.36

Table 11: 1500-1800 PWM step response comparison

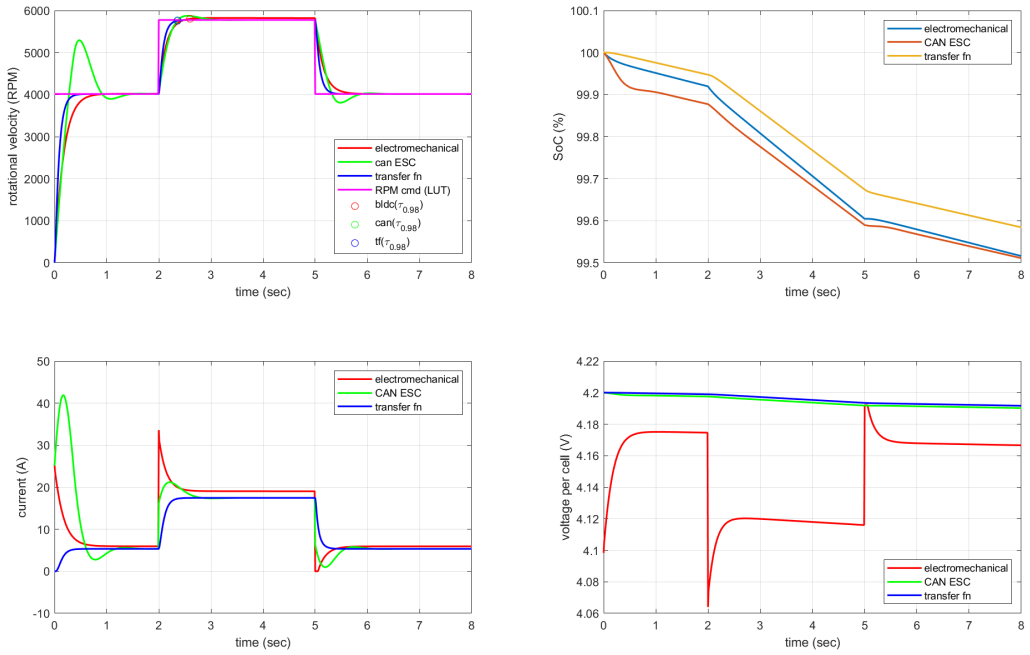


Figure 7: 1500-1800 PWM step response comparison

7.7 1700 + 50 PWM

Model	10-90% **rise time (s)**	**Settling time (s)**
electromechanical	0.28	2.51
CAN ESC	0.28	2.39
transfer fn	0.17	2.36

Table 12: 1700-1750 PWM step response comparison

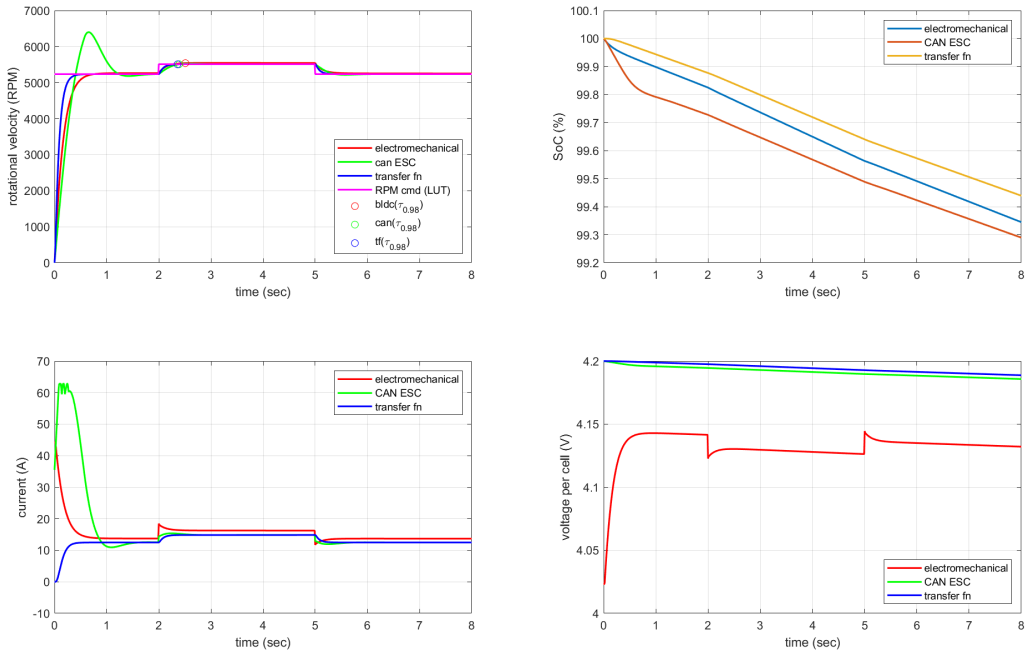


Figure 8: 1700-1750 PWM step response comparison

7.8 1700 + 150 PWM

Model	10-90% **rise time (s)**	**Settling time (s)**
electromechanical	0.28	2.54
CAN ESC	0.3	2.44
transfer fn	0.17	2.36

Table 13: 1700-1850 PWM step response comparison

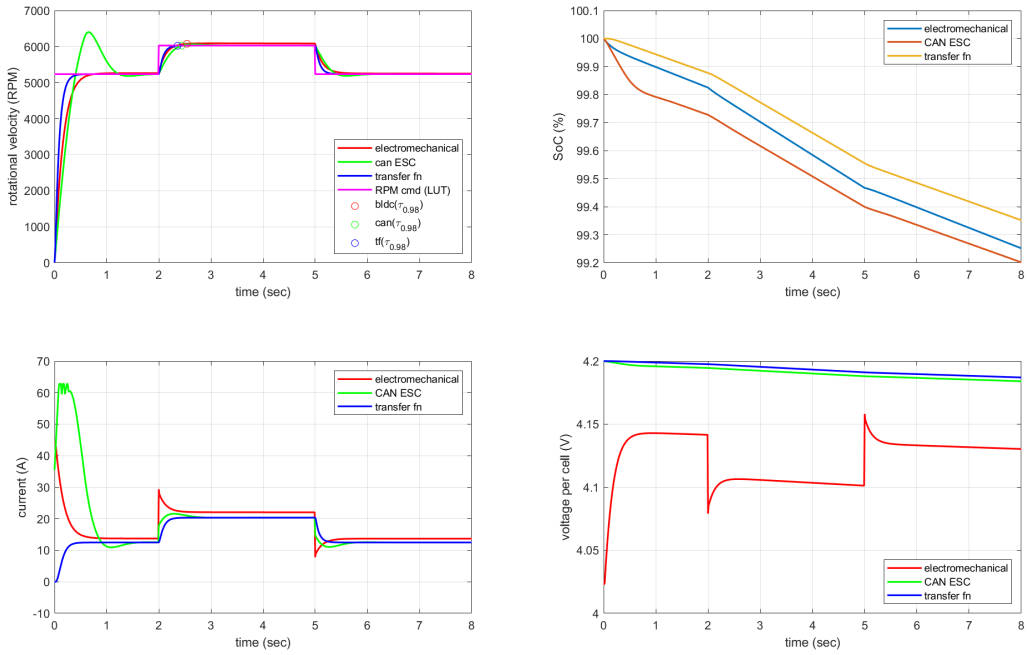


Figure 9: 1700-1850 PWM step response comparison

7.9 1700 + 300 PWM

Model	10-90% **rise time (s)**	**Settling time (s)**
electromechanical	0.27	2.53
CAN ESC	0.34	2.52
transfer fn	0.17	2.36

Table 14: 1700-2000 PWM step response comparison

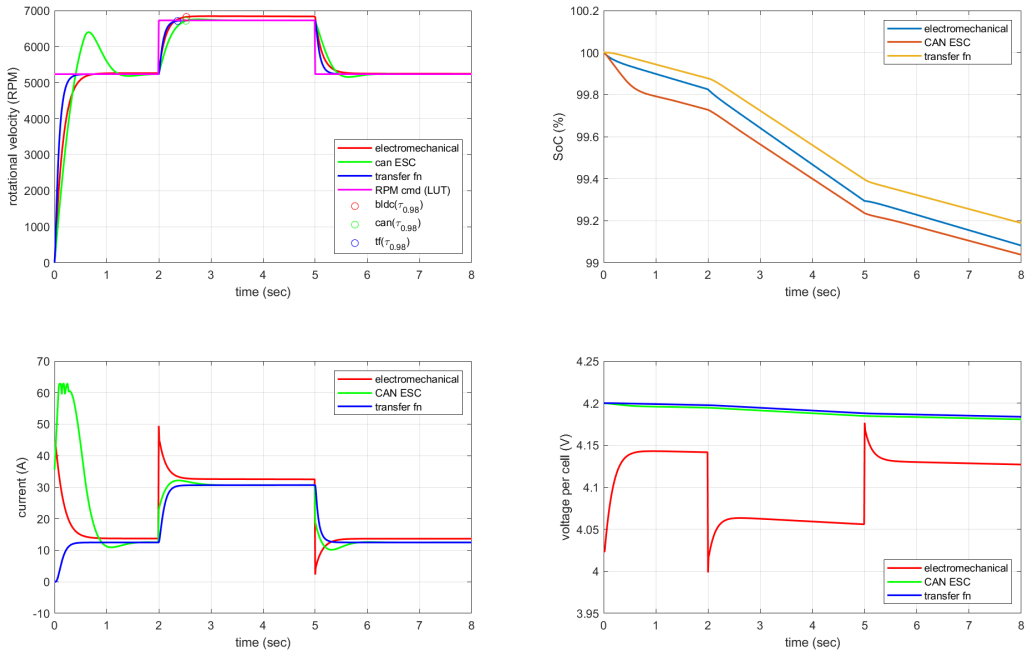


Figure 10: 1700-2000 PWM step response comparison

7.10 1000 + 1000 PWM

Model	10-90% **rise time (s)**	**Settling time (s)**
electromechanical	0.31	2.63
CAN ESC	0.49	2.69
transfer fn	0.17	2.36

Table 15: 1000-2000 PWM step response comparison

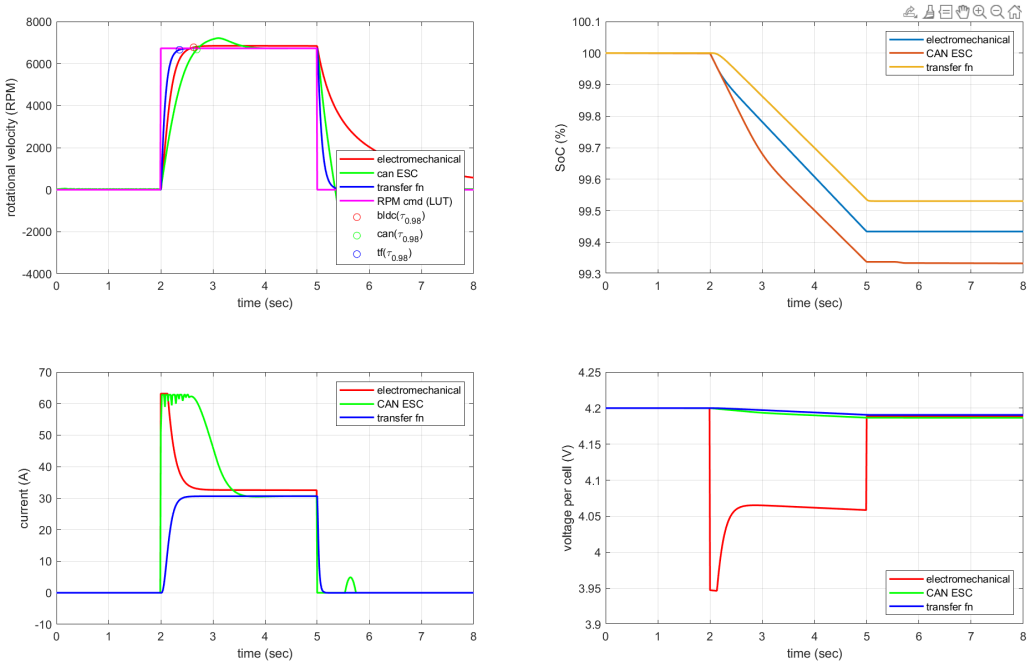


Figure 11: 1000-2000 PWM step response comparison

8 Sinusoidal and ramp inputs

8.1 Sinusoidal input

Qualitative description: Phase lag occurs first in the slower models, which are the electromechanical model and the CAN ESC model tuned to match. As known from the time constants reported in the previous sections, the transfer function model is faster than the other two models, as its cutoff frequency is higher.

Interesting details of model frequency are present as in the previous section. As voltage sag is properly captured in the electromechanical model, oscillations and attenuation are visible there.

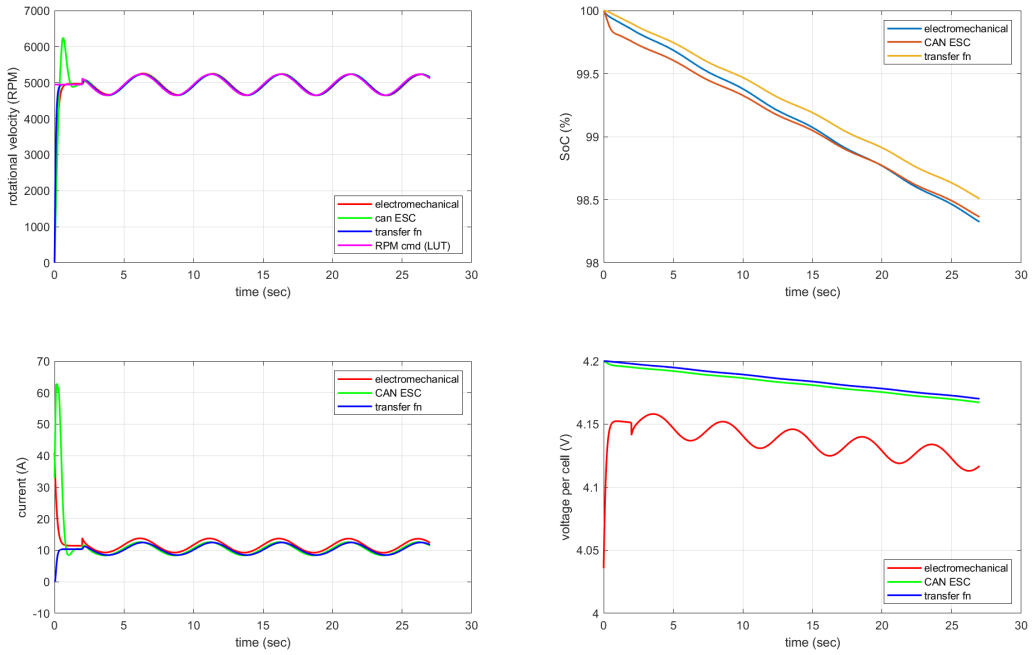


Figure 12: 0.2 Hz sine sweep results @ 1650 pwm

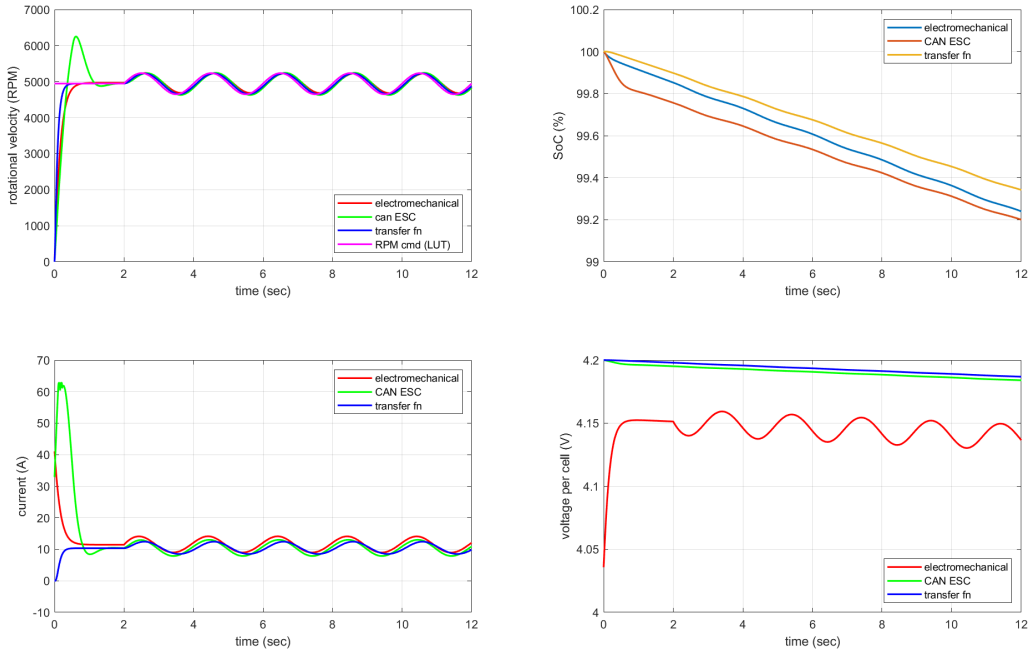


Figure 13: 0.5 Hz sine sweep results @ 1650 pwm

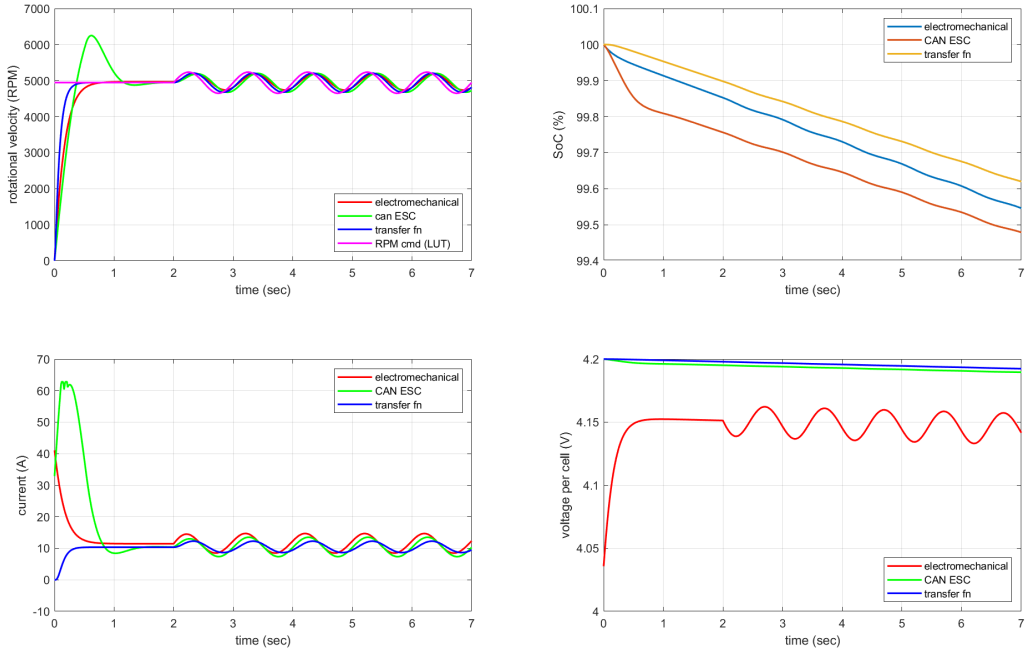


Figure 14: 1 Hz sine sweep results @ 1650 pwm

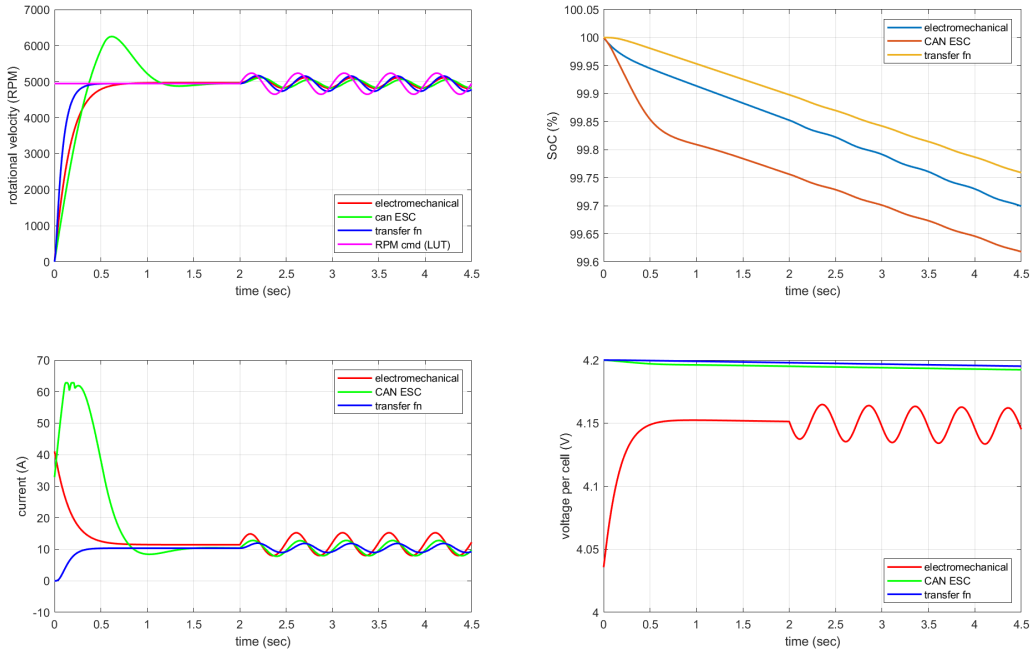


Figure 15: 2 Hz sine sweep results @ 1650 pwm

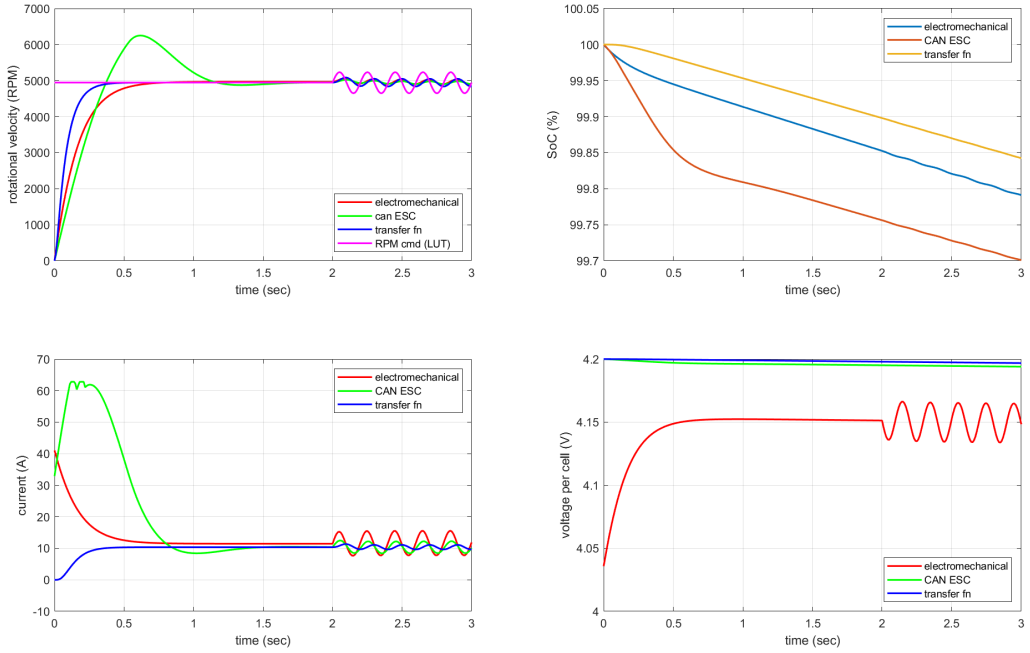


Figure 16: 5 Hz sine sweep results @ 1650 pwm

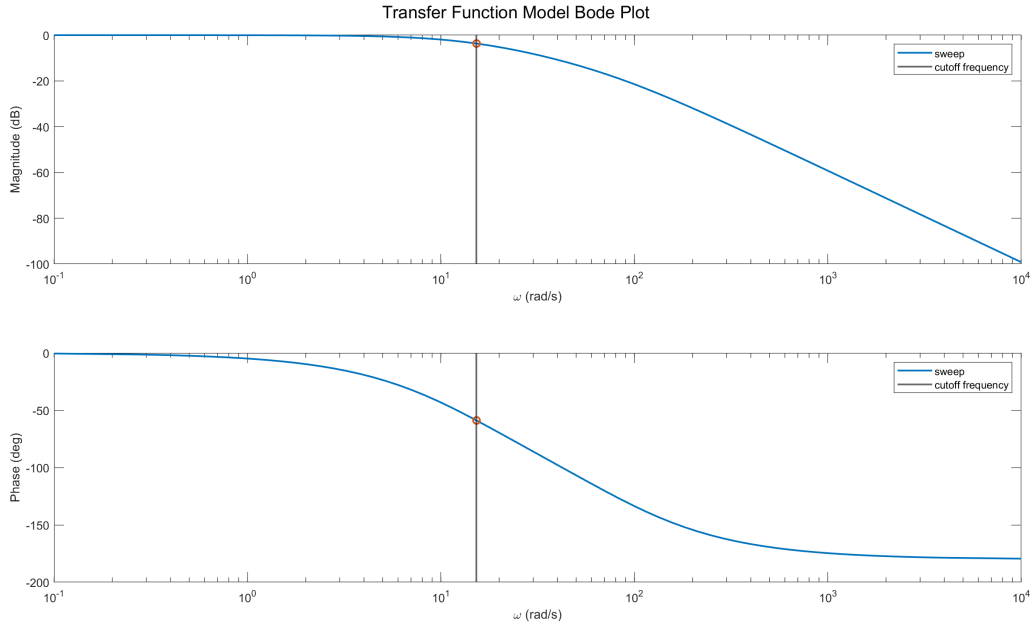


Figure 17: Transfer function model bode plot

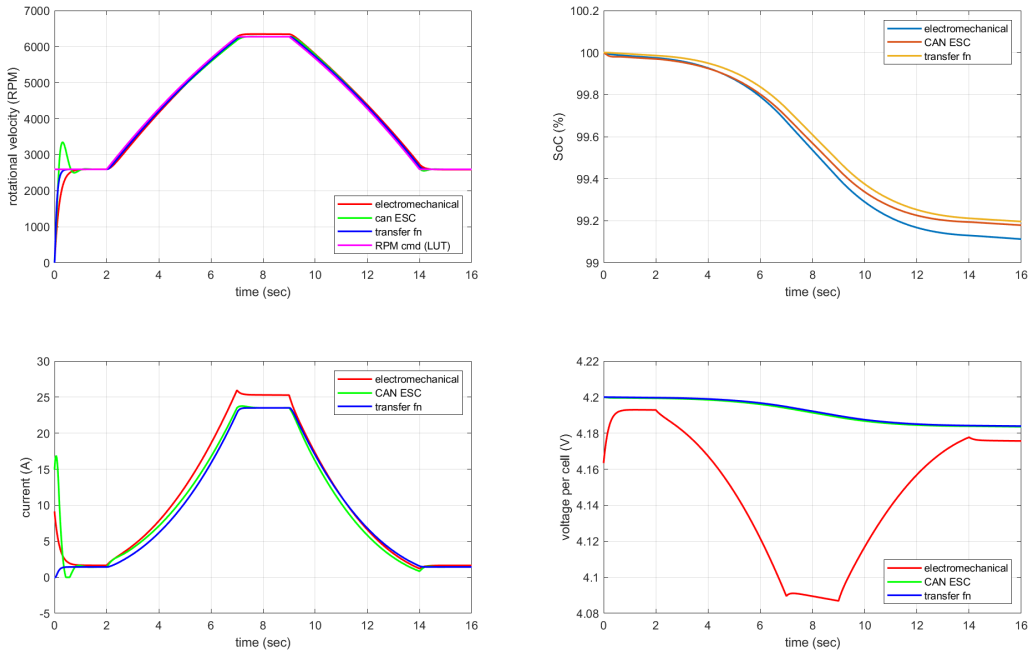


Figure 18: Ramp input response

Parameter	**Value**
Cutoff frequency (Hz)	2.43
Cutoff frequency (rad/s)	15.28

Table 16: Caption

8.2 Ramp input

Tracking accuracy is the same in magnitude throughout the ramp, but the sign changes to reflect the change of sign of the ramp. For example the CAN ESC model under predicts the instantaneous RPM during ramp up, but over predicts during ramp down.

9 Battery Discharge and Inflow Tests

9.1 Battery Discharge

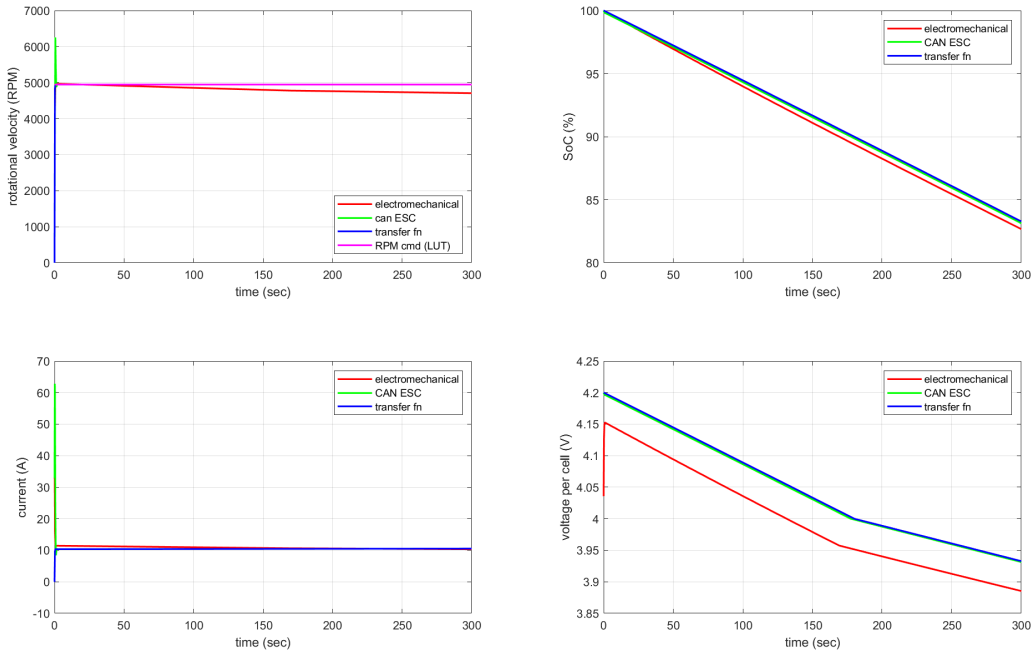


Figure 19: Model battery depletion test

Model	ΔRPM	ΔSoC (%)	$\Delta Current$ (A)	$\Delta Cell\ Potential$ (V)
Electromechanical	257.79	17.6	1.09	0.26
CAN ESC	0	16.51	-0.18	0.26
transfer function	0	16.51	-0.18	0.26

Table 17: Changes in parameters by model type

The modeled battery type is the same for the two lower fidelity models, explaining the same state of charge depletion and change in battery voltage. The electromechanical model loses slightly more of its battery percentage because the required power draw at a certain battery voltage varies with battery voltage, rather than staying fixed at the nominal value. For the state of charge of this test, above the nominal voltage, the power consumption is going to be higher.

9.2 Inflow and propeller torque

9.2.1 1300 PWM + 0 m/s Inflow

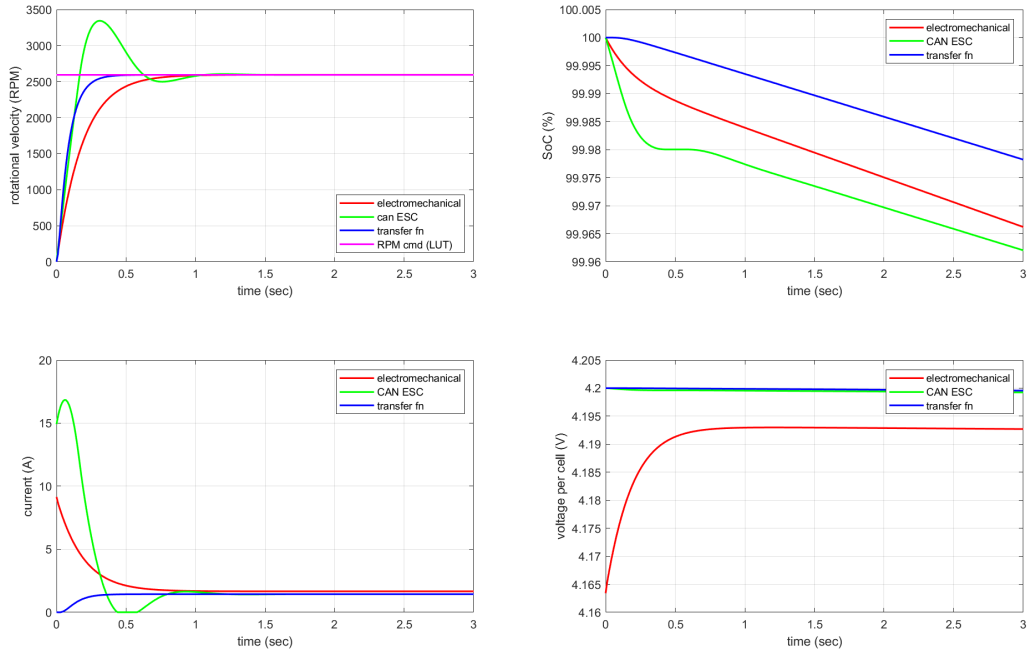


Figure 20: Standard plots

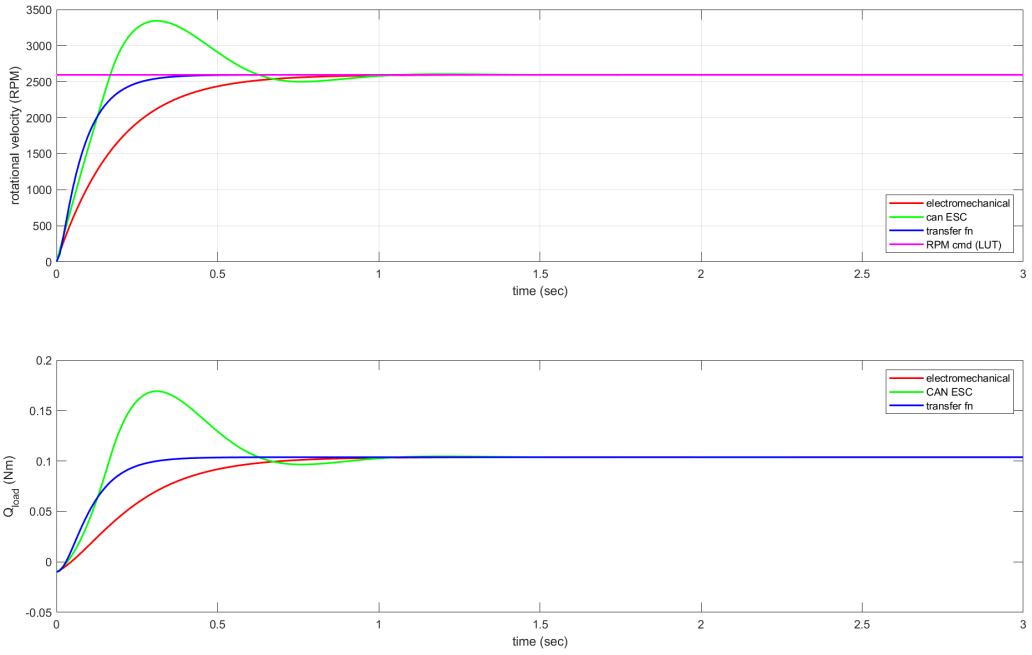


Figure 21: RPM and load torque

Model	Average no load torque (Nm)
Electromechanical	0.1083
CAN ESC	0.1083
Transfer function	0.1083

Table 18: Load torque across models

9.2.2 1300 PWM + 2.5 m/s Inflow

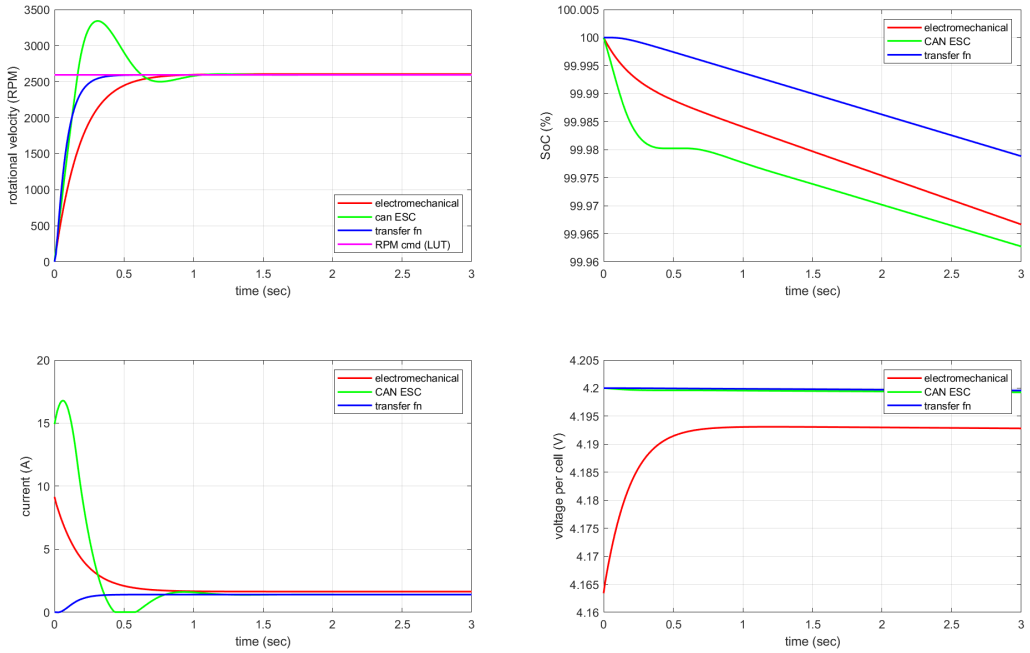


Figure 22: Standard plots

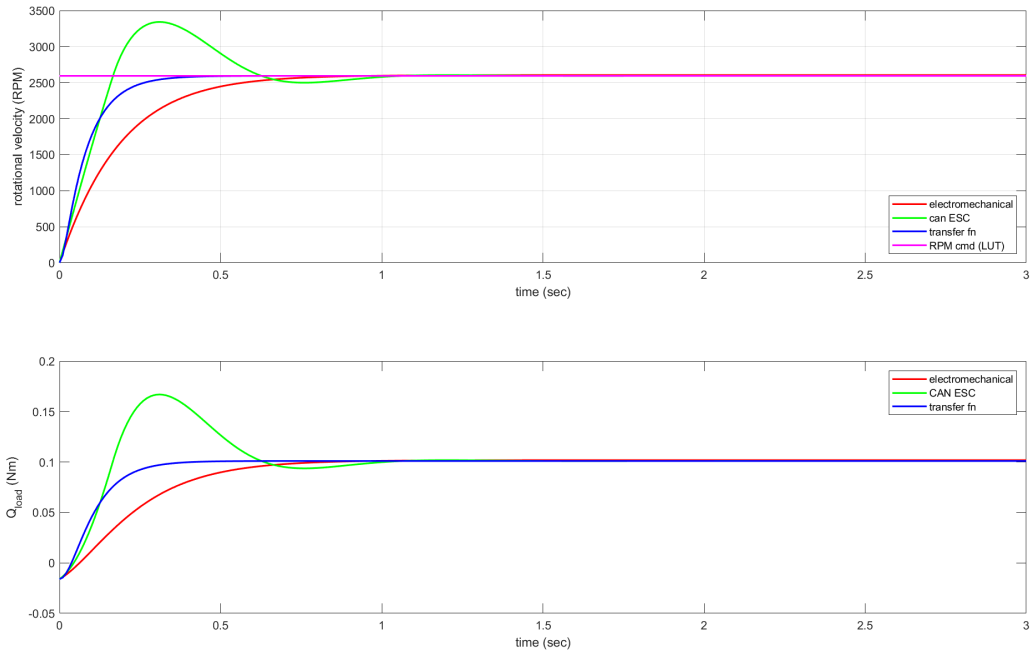


Figure 23: RPM and load torque

Model	Average no load torque (Nm)
Electromechanical	0.1018
CAN ESC	0.1010
Transfer function	0.1010

Table 19: Load torque across models

9.2.3 1300 PWM + 5.0 m/s Inflow

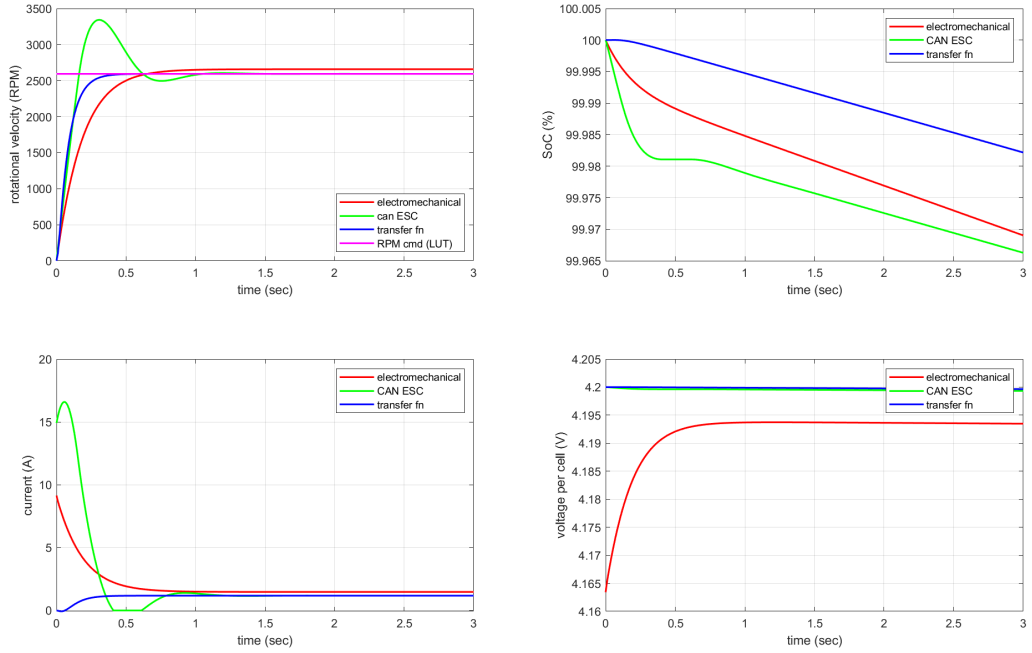


Figure 24: Standard plots

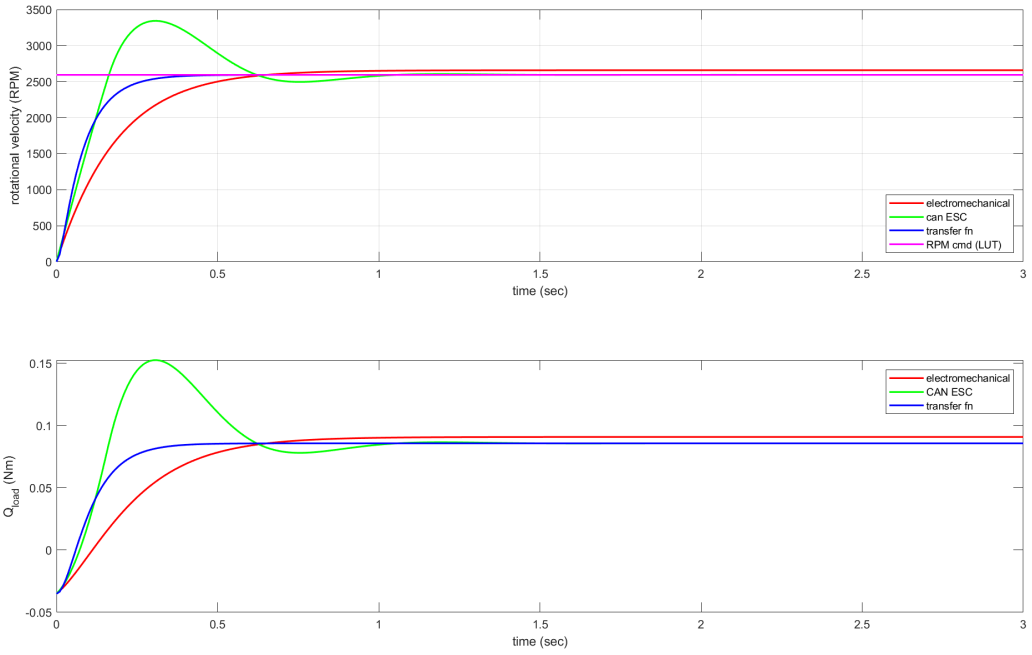


Figure 25: RPM and load torque

Model	Average no load torque (Nm)
Electromechanical	0.0908
CAN ESC	0.0857
Transfer function	0.0857

Table 20: Load torque across models

9.2.4 1500 PWM + 0 m/s Inflow

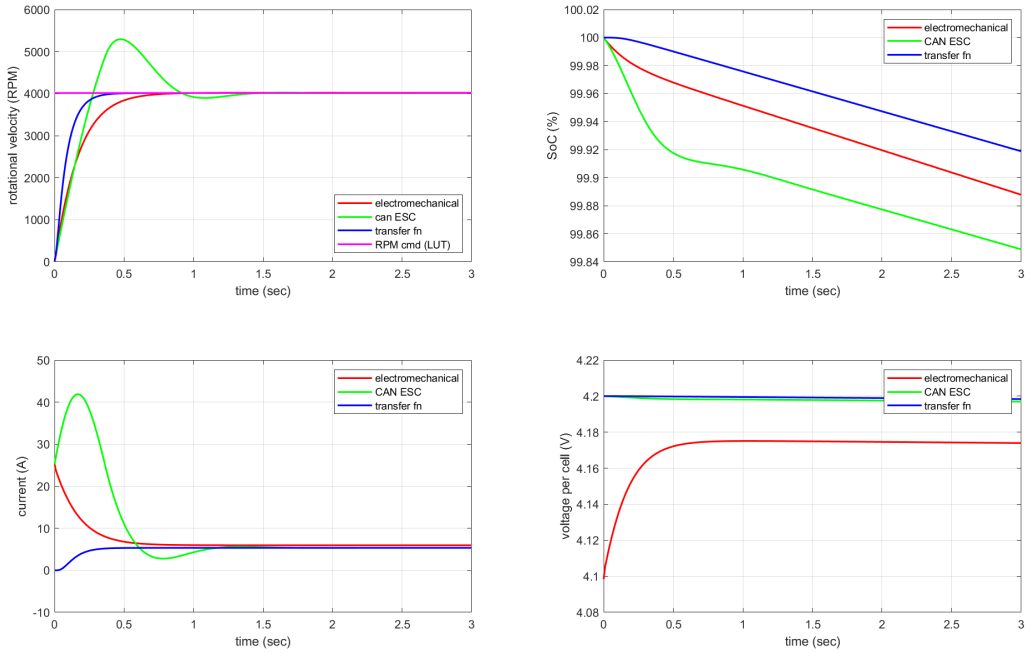


Figure 26: Standard plots

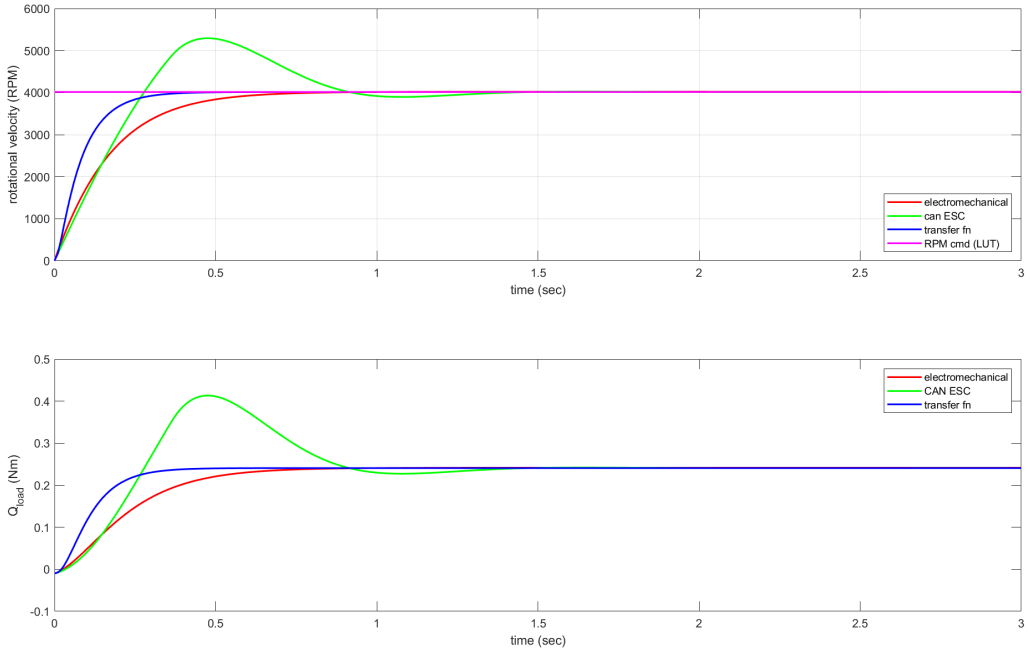


Figure 27: RPM and load torque

Model	Average no load torque (Nm)
Electromechanical	0.2415
CAN ESC	0.2408
Transfer function	0.2408

Table 21: Load torque across models

9.3 1500 PWM + 2.5 m/s Inflow

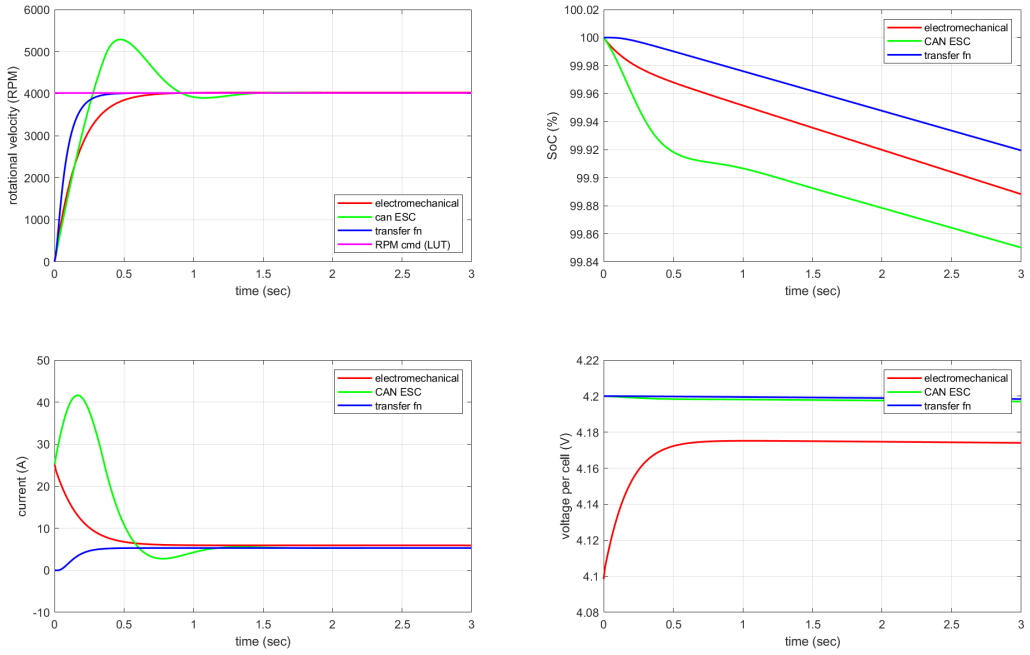


Figure 28: Standard plots

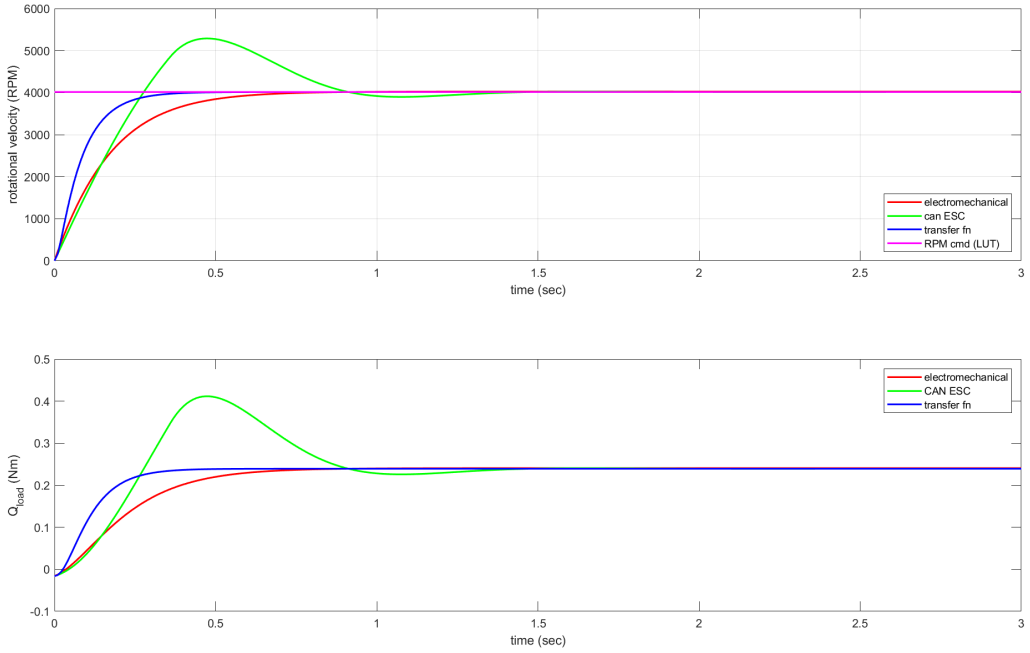


Figure 29: RPM and load torque

Model	Average no load torque (Nm)
Electromechanical	0.2405
CAN ESC	0.2392
Transfer function	0.2392

Table 22: Load torque across models

9.3.1 1500 PWM + 5.0 m/s Inflow

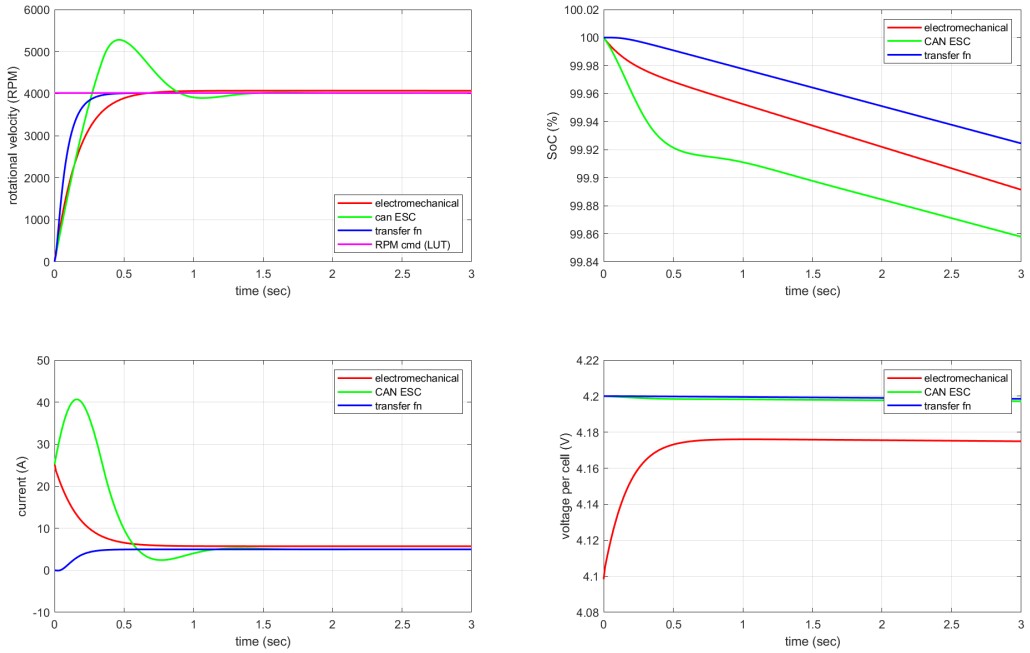


Figure 30: Standard plots

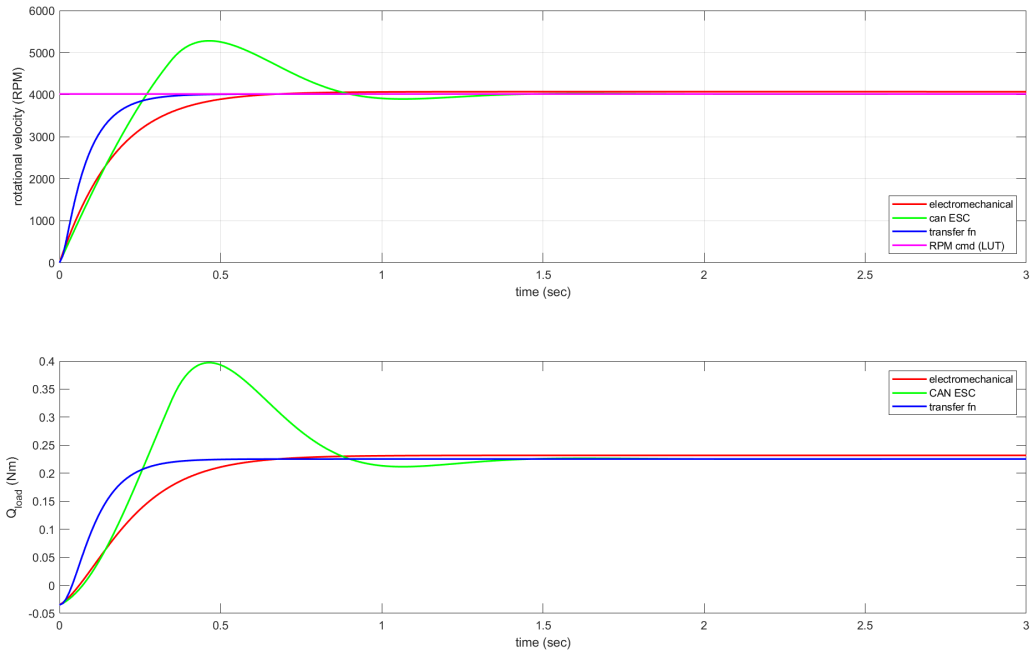


Figure 31: RPM and load torque

Model	Average no load torque (Nm)
Electromechanical	0.2318
CAN ESC	0.2253
Transfer function	0.2253

Table 23: Load torque across models

9.3.2 1700 PWM + 0 m/s Inflow

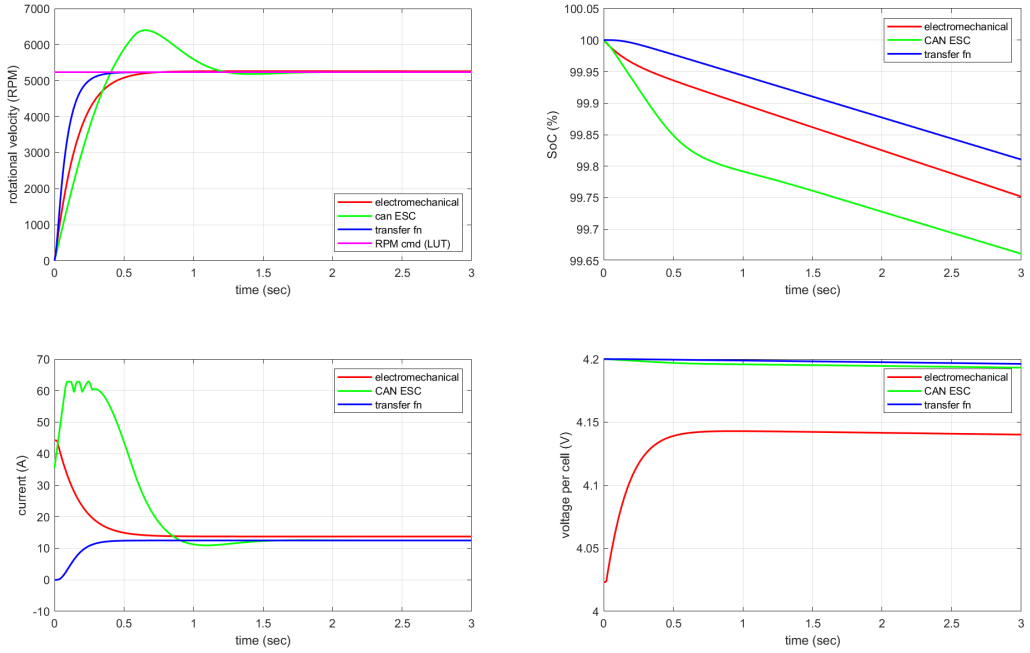


Figure 32: Standard plots

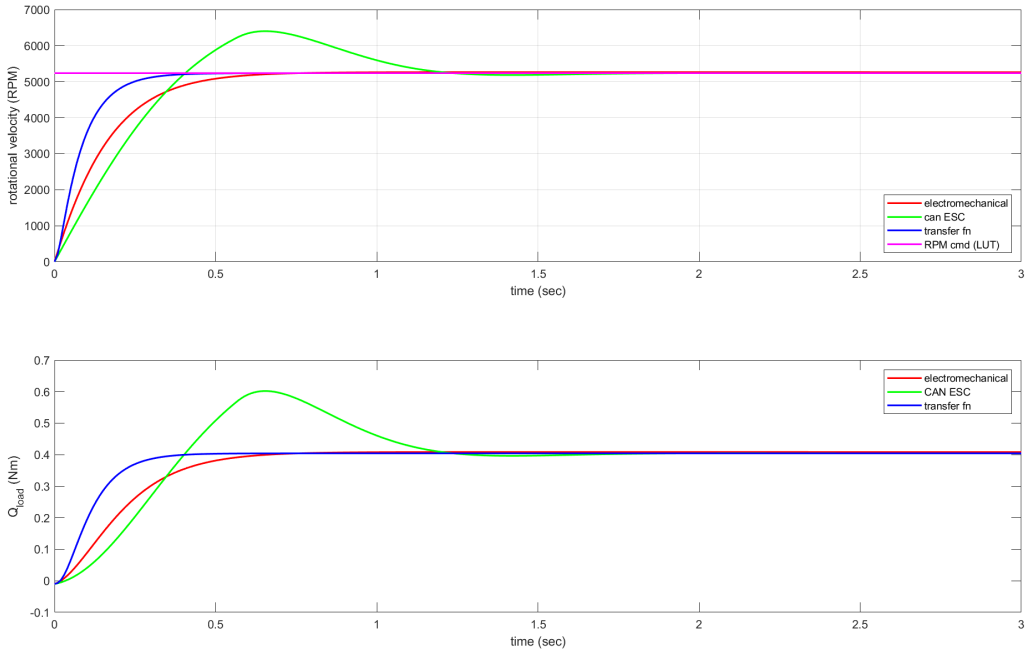


Figure 33: RPM and load torque

Model	Average no load torque (Nm)
Electromechanical	0.4082
CAN ESC	0.4082
Transfer function	0.4043

Table 24: Load torque across models

9.3.3 1700 PWM + 2.5 m/s Inflow

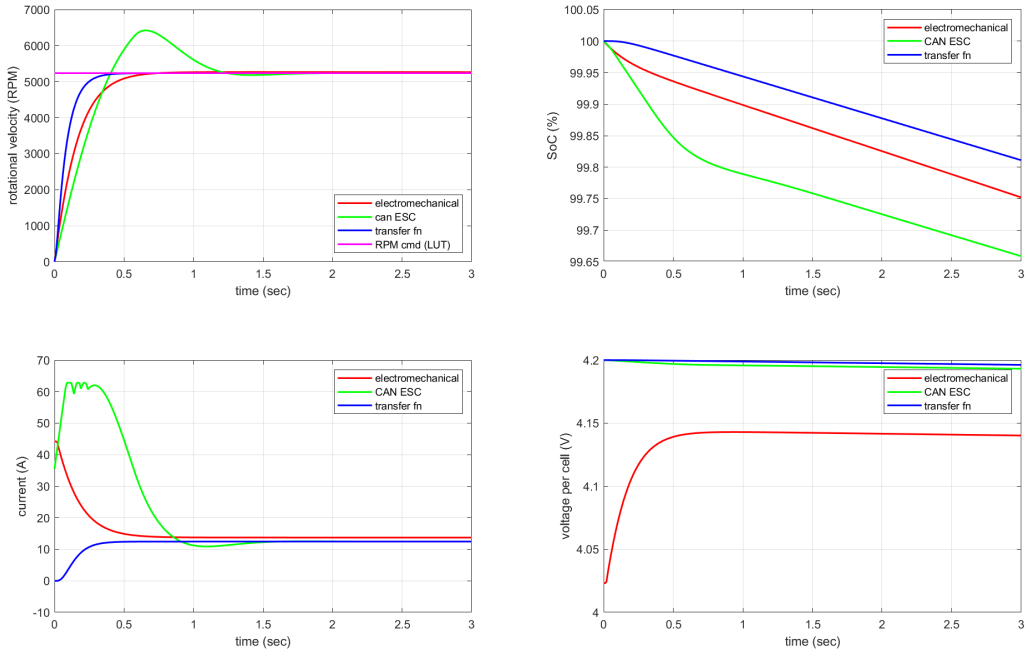


Figure 34: Standard plots

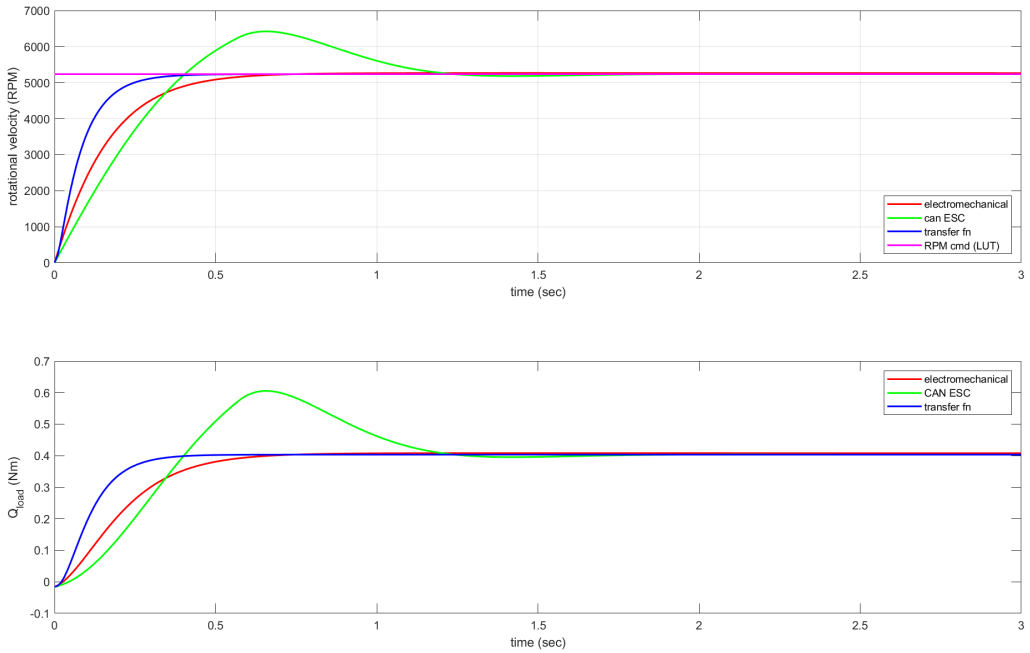


Figure 35: RPM and load torque

Model	Average no load torque (Nm)
Electromechanical	0.4077
CAN ESC	0.4034
Transfer function	0.4034

Table 25: Load torque across models

9.3.4 1700 PWM + 5.0 m/s Inflow

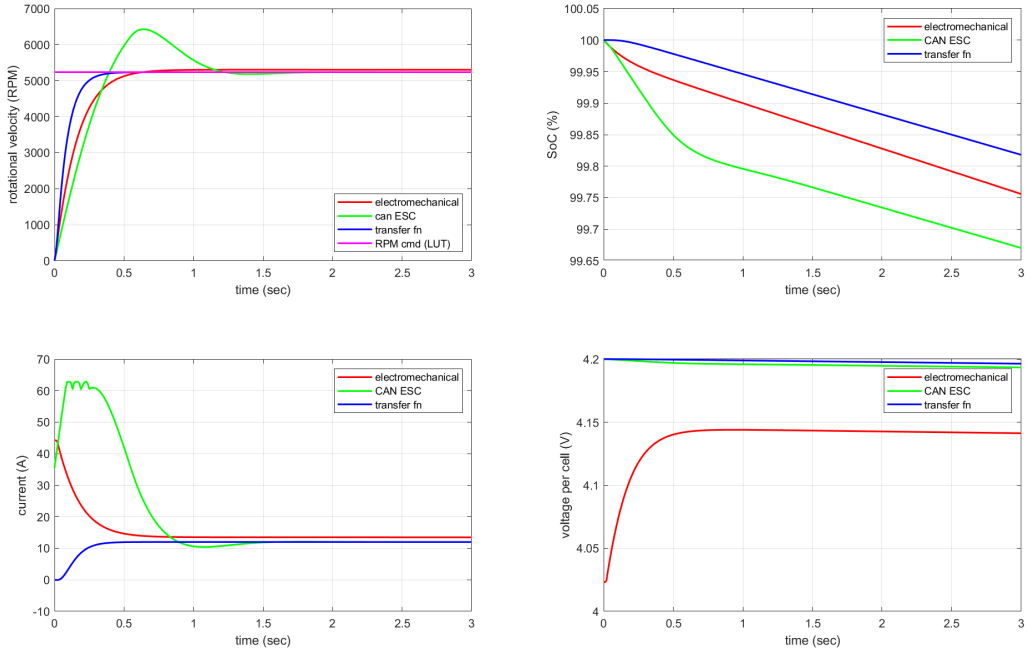


Figure 36: Standard plots

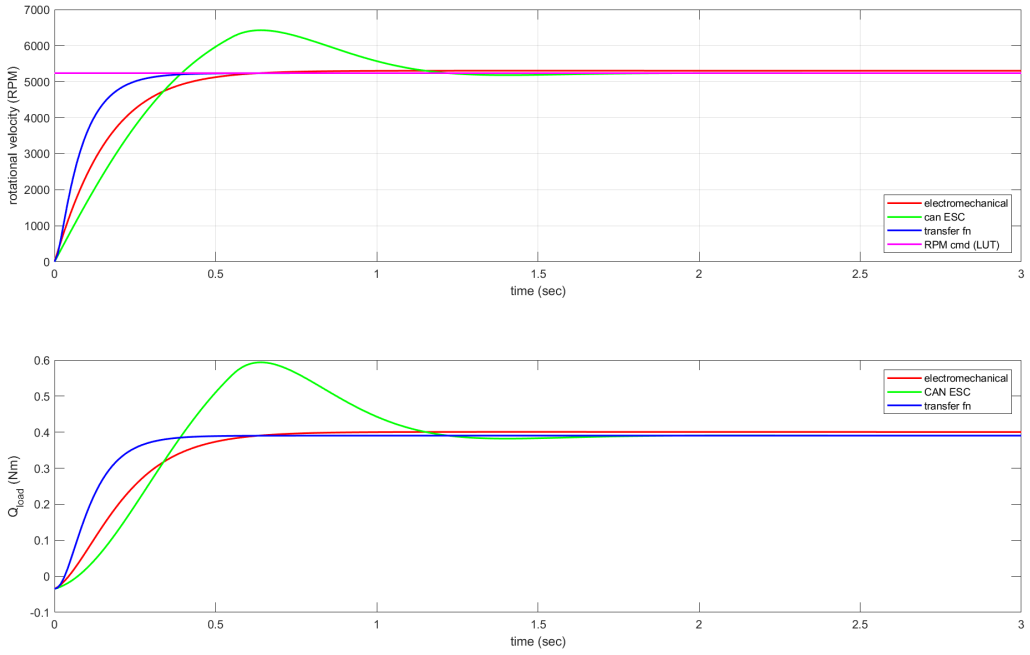


Figure 37: RPM and load torque

Model	Average no load torque (Nm)
Electromechanical	0.4004
CAN ESC	0.3903
Transfer function	0.3904

Table 26: Load torque across models

10 Discussion

1. How do rise time and settling time change with starting PWM and step size? Which cases suggest limiting, and what can the available plots establish

A. Rise time and settling time increase as the step size grows but stay relatively constant when starting PWM changes. The large step from 1000-2000 suggests limiting, as established from the saturation of battery current in the plots.

2. How does the oscillatory response change as frequency approaches and exceeds the actuator bandwidth? Relate your time histories to the provided transfer function.

A. As each actuator approaches its cutoff frequency (roughly 15.28 Hz 16), the signal becomes out of phase and "fights" the input. This results in attenuation of the input, as the actuator does not have enough time to achieve a commanded speed before the command changes again.

3. Why can the models predict different RPM trends during the constant-PWM battery-discharge test? Discuss the modeled connections between battery condition and motor speed.

A. Both the simplified and CAN ESC models close a loop over the commanded RPM, so the steady state RPM output from a constant RPM (fit one-to-one with a constant PWM command) will never change. The electromechanical model on the other hand is open-loop. With no feedback increasing the power from the battery, the motor will naturally slow down as the power across the ESC decreases proportionally to the battery SoC.

4. How does inflow change load torque at fixed PWM, and what role does the resulting RPM play?

A. Inflow reduces the load torque at a fixed RPM, as momentum is transferred less efficiently from the air to the propeller. The resulting RPM doesn't change in the case of the closed loop models, rather the amount of power drawn from the battery decreases. In the case of the open-loop model, the power draw from the battery remains constant, and the motor spins slightly faster (but does NOT produce more thrust).

5. What conditions does your PWM-to-RPM lookup represent? Explain why its relationship may change with battery condition or inflow.

A. The lookup table represents a the relationship with a fully charged battery at 0 inflow. At lower battery voltage, the steady state RPM would be lower, as there is less electrical power going through the ESC and motor at every PWM value. At higher inflow, the steady state RPM would increase, as there is less load torque from the air to balance the electrical power output from the ESC.

6. Which model would you select for a particular simulation task? Explain the relevant tradeoff between model complexity and the behavior represented.

A. I would select the electromechanical model when I need to capture high fidelity behavior of a propulsion system with a feed-forward ESC. I would also use its battery model when voltage sag during dynamic maneuvers are likely to change simulation predictions of SoC. I would select the CAN ESC model if I needed to close the loop over PWM, but still wanted to retain a high fidelity model of the system dynamics. This model would be ideal for a subscale helicopter that flies with varying collective at constant RPM. Finally, I may select a 1st or 2nd order transfer function of the actuator for control law design, as the linear system makes incorporating actuator dynamics tractable.